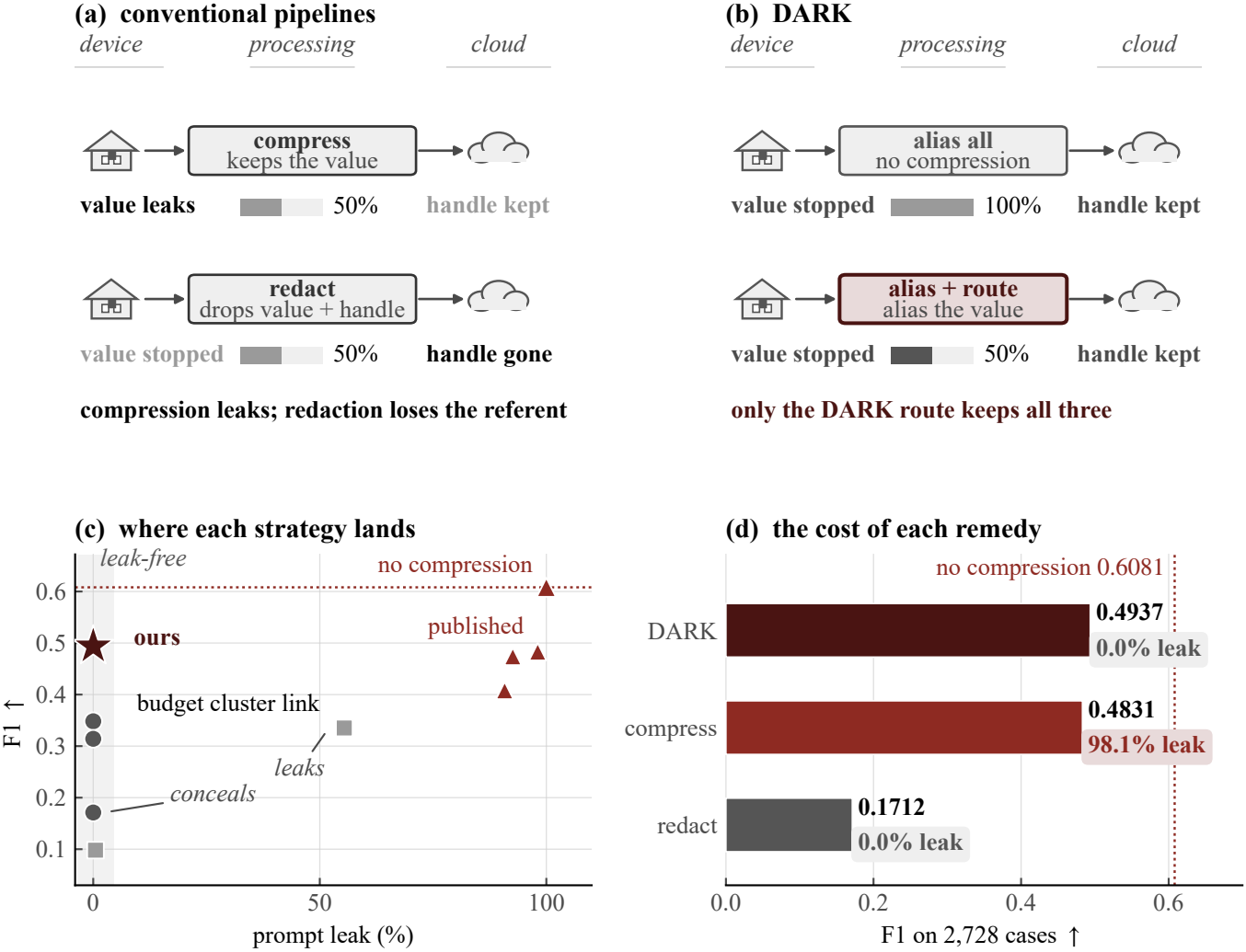

(a) conventional pipelines
device
processing
cloud
compress
keeps the value
value leaks
50%
handle kept
redact
drops value + handle
value stopped
50%
handle gone
compression leaks; redaction loses the referent
(b) DARK
device
processing
cloud
alias all
no compression
value stopped
100%
handle kept
alias + route
alias the value
value stopped
50%
handle kept
only the DARK route keeps all three
(c) where each strategy lands
leak-free
no compression
published
ours
budget cluster link
leaks
conceals
F1 ↑
prompt leak (%)
(d) the cost of each remedy
no compression 0.6081
DARK
0.4937
0.0% leak
compress
0.4831
98.1% leak
redact
0.1712
0.0% leak
F1 on 2,728 cases ↑